Harouna Thiam

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EDUCATION

Duke University Durham, NC

B.S. in Computer Science Expected May 2028

Relevant Coursework: DSA, Computer Systems, Discrete Math, Probability, Linear Algebra, Computer Network Architecture

Awards & Fellowships: **HackDuke** '25 Winner, **U.S. Congressional** App Challenge Winner, **Amazon** Future Engineer Scholarship (1/400 selected), **LinkedIn** Possibilities Scholarship (1/25 selected), **Jane Street** UNBOXED Scholarship (1/38 selected), **Walmart Global Tech** SWE Fellow (<1.5% selected), **MLT** Career Prep Fellow, **Goldman Sachs** Engineering Summit Fellow

SKILLS

Languages: Go, Python, Java, TypeScript, R, HTML, CSS, SQL, C

Infrastructure: AWS (Lambda, S3, Bedrock, SNS, DynamoDB, EKS), Docker, git, CI/CD, REST, Kubernetes, JUnit, MySQL, kind

AI/ML/Data: LangChain, Weaviate, Pandas, Prompt Engineering, NumPy, Tensorflow, Codex, Claude Code, Cursor

EXPERIENCE

Software Engineer Intern, ***Amazon Web Services*** May. 2026 – Aug. 2026

- Architected a highly-available notification scheduler in **Go** that monitors cert. authority changes in **600k+** Kubernetes clusters
- Deployed serverless architecture to route expiry messages with AWS (Lambda, EB) to mitigate TLS failure in **9,900+** companies
- Developed a serverless async reconciliation loop to redrive failures in a state machine, achieving **99%** message send rates
- Implemented an atomic transaction model for writes to DynamoDB, preventing desyncs between CA operations and notifications
- Wired notification coverage for all AWS partitions, implementing DNS lookup for on-deployment endpoint building & integrations

Software Engineer Intern, ***Amazon*** May. 2025 – Aug. 2025

- Built and owned serverless cloud LLM pipeline using Java and AWS to automate data augmentation and validation for Fire Tablets
- Reduced JSON processing time by **87%** by using a distributed parallel map state with AWS Bedrock and AWS Step Functions
- Designed a callback token mechanism with TypeScript SNS listeners and Lambda gateways to synchronize validation responses
- Implemented an Utterance Validation Framework with handoffs to S3 between workflow states to validate Alexa text-to-speech
- Ensured code quality with automated testing suites (unit, integration, load) with JUnit, integrated into a CI/CD pipeline

Undergraduate Researcher, ***Duke Systems Group*** Sept. 2026 – Current

- Investigating transparent GPU utilization sharing in Duke Compute Cluster (DCC) GPU partitions under Dr. Matthew Lentz
- Utilizing CPU preprofiling to simulate job memory usage (CUDACachingAllocator) to determine job placement in GPU clusters

Software Engineer, ***Duke Fuqua School of Business*** Apr. 2025 – Mar. 2026

- Architected and deployed a proprietary Excel autograder (Python, Openpyxl, Pandas) for **400+** student course at Duke Fuqua
- Decreased grading time for TAs by **42%** on average with the deployed software, managing an 8-person TA team
- Integrated generative AI prompts across **150+** pages of textbooks to create AI-assisted instructions for graduate coursework
- Facilitated the integration of a new Learning Management System (LMS) for AI-Enhanced graduate student learning

Software Engineer Intern, ***ZSuite Technologies*** Jun. 2023 – Aug. 2023

- Developed LLM chatbot tool with Python and Langchain, resulting in increased efficiency of company representative response
- Curated relevant context to an LLM from CSV files through a vector store with Weaviate and RAG, improving query accuracy
- Containerized code and workspace with Docker and Jupyter Notebooks to create a scalable and modular codebase

PROJECTS

Reaper: Kubernetes Controller for Automatic Gang Protection in ML Training Clusters May. 2026

- Built a Kubernetes controller in **Go** automating GPU node lifecycles, automating cluster state changes with reconciler for EKS
- Designed an anti-disruption policy layer to automatically protect GPU node gang schedules from deprovisionment on Karpenter
- Validated end-to-end against a multi-node kind cluster with simulated GPU node pools and distributed training job manifests
- Designed first-fit bin-packing logic to detect consolidation candidates, flagging underutilized GPU nodes for Karpenter rightsizing

Duke University 'Code for Good' Hackathon (Odyssey) • ***1st Place App in Sustainability*** Feb. 2025

- Developed a full mobile application in a team setting at HackDuke using Flutter and Gemini, securing **150+** downloads

United States Congressional App Challenge (Florida GreenGuard AI) • ***1st Place App - Florida*** Nov. 2023

- Integrated image classification models using TensorFlow, Kotlin, and OpenAI to detect plant invasive species and ailments

PROFESSIONAL DEVELOPMENT

Alliance for Innovative and Interdisciplinary Computing Education (AiiCE) • *National Student Advisory Board Member*
Cornell Tech • *Break Through Tech AI Program Fellow*
LinkedIn • *Possibilities [in] Technology Cohort Member*